

```

In [3]: import tensorflow as tf
from tensorflow.keras.datasets import cifar10
from tensorflow.keras.models import Sequential
from tensorflow.keras.layers import Flatten, Dense
from tensorflow.keras.utils import to_categorical
import matplotlib.pyplot as plt

# 1. Load CIFAR-10 dataset
(x_train, y_train), (x_test, y_test) = cifar10.load_data()

# 2. Normalize images
x_train = x_train / 255.0
x_test = x_test / 255.0

# 3. One-hot encode labels
y_train = to_categorical(y_train, 10)
y_test = to_categorical(y_test, 10)

# 4. CIFAR-10 class names
class_names = [
    'Airplane', 'Automobile', 'Bird', 'Cat', 'Deer',
    'Dog', 'Frog', 'Horse', 'Ship', 'Truck'
]

# 5. Create FFNN with THREE hidden layers
def create_model(units, activation):
    model = Sequential([
        Flatten(input_shape=(32, 32, 3)),

        Dense(units[0], activation=activation),
        Dense(units[1], activation=activation),
        Dense(units[2], activation=activation),

        Dense(10, activation='softmax')
    ])

    model.compile(
        optimizer='adam',
        loss='categorical_crossentropy',
        metrics=['accuracy']
    )

    return model

# 6. Hyperparameters
hidden_units_list = [
    (1024, 512, 256),
    (512, 256, 128),
    (256, 128, 64),
    (128, 64, 32)
]

# Test ALL three activation functions
activation_list = ['relu', 'tanh', 'sigmoid']

# 7. Store results
results = []

best_accuracy = 0
best_model = None
best_config = None

# 8. Train all configurations
model_number = 1

for units in hidden_units_list:
    for activation in activation_list:

```

```

print("\n=====")
print("Training Model", model_number)
print("Hidden Layers:", units)
print("Activation:", activation)
print("=====")

model = create_model(units, activation)

# Train model
model.fit(
    x_train,
    y_train,
    epochs=5,
    batch_size=64,
    verbose=1
)

# Evaluate model
loss, accuracy = model.evaluate(
    x_test,
    y_test,
    verbose=0
)

print("Test Loss:", loss)
print("Test Accuracy:", accuracy)

# Store results
results.append({
    "Model": model_number,
    "Hidden Layers": units,
    "Activation": activation,
    "Test Loss": loss,
    "Test Accuracy": accuracy
})

# Find best model
if accuracy > best_accuracy:
    best_accuracy = accuracy
    best_model = model
    best_config = (units, activation)

model_number += 1

# 9. Display ALL model configurations
print("\n\n=====")
print("          CONFIGURATION OF ALL MODELS")
print("=====")

for result in results:
    print("\nModel", result["Model"])
    print("Hidden Layers :", result["Hidden Layers"])
    print("Activation      :", result["Activation"])
    print("Test Loss        :", round(result["Test Loss"], 4))
    print("Test Accuracy    :", round(result["Test Accuracy"], 4))

# 10. Display best model
print("\n\n=====")
print("          BEST MODEL")
print("=====")

print("Hidden Layers :", best_config[0])
print("Activation      :", best_config[1])
print("Best Accuracy   :", round(best_accuracy, 4))

# 11. Predict one test image
index = 0

prediction = best_model.predict(
    x_test[index:index+1],

```

```
        verbose=0
    )

    predicted_class = prediction.argmax()
    actual_class = y_test[index].argmax()

    print("\nPredicted Label:", class_names[predicted_class])
    print("Actual Label   :", class_names[actual_class])

# 12. Display the image
plt.imshow(x_test[index])
plt.title(
    "Predicted: " + class_names[predicted_class] +
    " | Actual: " + class_names[actual_class]
)
plt.axis('off')
plt.show()
```

=====

Training Model 1

Hidden Layers: (1024, 512, 256)

Activation: relu

=====

Epoch 1/5

782/782  25s 30ms/step - accuracy: 0.3187 - loss: 1.8791

Epoch 2/5

782/782  24s 31ms/step - accuracy: 0.4064 - loss: 1.6558

Epoch 3/5

782/782  24s 30ms/step - accuracy: 0.4384 - loss: 1.5671

Epoch 4/5

782/782  23s 30ms/step - accuracy: 0.4578 - loss: 1.5115

Epoch 5/5

782/782  25s 32ms/step - accuracy: 0.4781 - loss: 1.4631

Test Loss: 1.48203706741333

Test Accuracy: 0.47609999775886536

=====

Training Model 2

Hidden Layers: (1024, 512, 256)

Activation: tanh

=====

Epoch 1/5

782/782  25s 32ms/step - accuracy: 0.2297 - loss: 2.0428

Epoch 2/5

782/782  24s 31ms/step - accuracy: 0.2784 - loss: 1.9428

Epoch 3/5

782/782  24s 30ms/step - accuracy: 0.2930 - loss: 1.9160

Epoch 4/5

782/782  23s 30ms/step - accuracy: 0.2921 - loss: 1.9130

Epoch 5/5

782/782  24s 31ms/step - accuracy: 0.3016 - loss: 1.8962

Test Loss: 1.9083292484283447

Test Accuracy: 0.319599986076355

=====

Training Model 3

Hidden Layers: (1024, 512, 256)

Activation: sigmoid

=====

Epoch 1/5

782/782  24s 30ms/step - accuracy: 0.2916 - loss: 1.9117

Epoch 2/5

782/782  23s 30ms/step - accuracy: 0.3696 - loss: 1.7412

Epoch 3/5

782/782  23s 30ms/step - accuracy: 0.3894 - loss: 1.6866

Epoch 4/5

782/782  25s 32ms/step - accuracy: 0.4039 - loss: 1.6516

Epoch 5/5

782/782  24s 31ms/step - accuracy: 0.4236 - loss: 1.6045

Test Loss: 1.6172733306884766

Test Accuracy: 0.41350001096725464

=====

Training Model 4

Hidden Layers: (512, 256, 128)

Activation: relu

=====

Epoch 1/5

782/782  10s 12ms/step - accuracy: 0.3180 - loss: 1.8819

Epoch 2/5

782/782  9s 11ms/step - accuracy: 0.3937 - loss: 1.6851

Epoch 3/5

782/782  10s 12ms/step - accuracy: 0.4293 - loss: 1.5928

Epoch 4/5

782/782  9s 11ms/step - accuracy: 0.4500 - loss: 1.5374

Epoch 5/5

782/782  10s 13ms/step - accuracy: 0.4672 - loss: 1.4900

Test Loss: 1.4871188402175903

Test Accuracy: 0.46470001339912415

=====

Training Model 5
Hidden Layers: (512, 256, 128)
Activation: tanh
=====

Epoch	782/782	Time	Step	Accuracy	Loss
Epoch 1/5		10s	11ms/step	0.2341	2.0366
Epoch 2/5		9s	12ms/step	0.2941	1.9115
Epoch 3/5		9s	12ms/step	0.3063	1.8811
Epoch 4/5		9s	12ms/step	0.3216	1.8495
Epoch 5/5		9s	12ms/step	0.3280	1.8297

Test Loss: 1.811009168624878
Test Accuracy: 0.3398000019073486

=====

Training Model 6
Hidden Layers: (512, 256, 128)
Activation: sigmoid
=====

Epoch	782/782	Time	Step	Accuracy	Loss
Epoch 1/5		10s	12ms/step	0.2812	1.9381
Epoch 2/5		9s	11ms/step	0.3574	1.7627
Epoch 3/5		10s	13ms/step	0.3809	1.7090
Epoch 4/5		10s	13ms/step	0.3974	1.6678
Epoch 5/5		9s	12ms/step	0.4121	1.6249

Test Loss: 1.6095858812332153
Test Accuracy: 0.42399999499320984

=====

Training Model 7
Hidden Layers: (256, 128, 64)
Activation: relu
=====

Epoch	782/782	Time	Step	Accuracy	Loss
Epoch 1/5		5s	5ms/step	0.3160	1.8843
Epoch 2/5		4s	5ms/step	0.3938	1.6828
Epoch 3/5		4s	5ms/step	0.4251	1.6019
Epoch 4/5		4s	6ms/step	0.4481	1.5491
Epoch 5/5		5s	6ms/step	0.4637	1.5027

Test Loss: 1.5528537034988403
Test Accuracy: 0.4462999999523163

=====

Training Model 8
Hidden Layers: (256, 128, 64)
Activation: tanh
=====

Epoch	782/782	Time	Step	Accuracy	Loss
Epoch 1/5		5s	6ms/step	0.2643	1.9852
Epoch 2/5		4s	5ms/step	0.3206	1.8598
Epoch 3/5		4s	5ms/step	0.3406	1.8204
Epoch 4/5		4s	5ms/step	0.3464	1.8106
Epoch 5/5		4s	6ms/step	0.3529	1.7967

Test Loss: 1.8037577867507935
Test Accuracy: 0.3587999939918518

=====

Training Model 9

Hidden Layers: (256, 128, 64)
Activation: sigmoid

=====

Epoch 1/5

782/782  5s 6ms/step - accuracy: 0.2782 - loss: 1.9648

Epoch 2/5

782/782  4s 6ms/step - accuracy: 0.3529 - loss: 1.7800

Epoch 3/5

782/782  4s 5ms/step - accuracy: 0.3753 - loss: 1.7182

Epoch 4/5

782/782  4s 5ms/step - accuracy: 0.3927 - loss: 1.6739

Epoch 5/5

782/782  4s 5ms/step - accuracy: 0.4054 - loss: 1.6412

Test Loss: 1.649930477142334

Test Accuracy: 0.3984000086784363

=====

Training Model 10

Hidden Layers: (128, 64, 32)

Activation: relu

=====

Epoch 1/5

782/782  3s 3ms/step - accuracy: 0.3004 - loss: 1.9221

Epoch 2/5

782/782  2s 3ms/step - accuracy: 0.3775 - loss: 1.7275

Epoch 3/5

782/782  2s 3ms/step - accuracy: 0.4059 - loss: 1.6548

Epoch 4/5

782/782  3s 4ms/step - accuracy: 0.4301 - loss: 1.5959

Epoch 5/5

782/782  3s 3ms/step - accuracy: 0.4444 - loss: 1.5551

Test Loss: 1.5425413846969604

Test Accuracy: 0.44940000772476196

=====

Training Model 11

Hidden Layers: (128, 64, 32)

Activation: tanh

=====

Epoch 1/5

782/782  3s 3ms/step - accuracy: 0.2761 - loss: 1.9745

Epoch 2/5

782/782  3s 4ms/step - accuracy: 0.3361 - loss: 1.8334

Epoch 3/5

782/782  3s 4ms/step - accuracy: 0.3461 - loss: 1.8026

Epoch 4/5

782/782  2s 3ms/step - accuracy: 0.3617 - loss: 1.7674

Epoch 5/5

782/782  3s 4ms/step - accuracy: 0.3701 - loss: 1.7484

Test Loss: 1.714001178741455

Test Accuracy: 0.38589999079704285

=====

Training Model 12

Hidden Layers: (128, 64, 32)

Activation: sigmoid

=====

Epoch 1/5

782/782  4s 4ms/step - accuracy: 0.2204 - loss: 2.0814

Epoch 2/5

782/782  4s 5ms/step - accuracy: 0.3174 - loss: 1.8765

Epoch 3/5

782/782  3s 4ms/step - accuracy: 0.3479 - loss: 1.7978

Epoch 4/5

782/782  3s 4ms/step - accuracy: 0.3668 - loss: 1.7466

Epoch 5/5

782/782  3s 4ms/step - accuracy: 0.3776 - loss: 1.7145

Test Loss: 1.7647556066513062

Test Accuracy: 0.36340001225471497

=====

CONFIGURATION OF ALL MODELS

=====

Model 1
Hidden Layers : (1024, 512, 256)
Activation : relu
Test Loss : 1.482
Test Accuracy : 0.4761

Model 2
Hidden Layers : (1024, 512, 256)
Activation : tanh
Test Loss : 1.9083
Test Accuracy : 0.3196

Model 3
Hidden Layers : (1024, 512, 256)
Activation : sigmoid
Test Loss : 1.6173
Test Accuracy : 0.4135

Model 4
Hidden Layers : (512, 256, 128)
Activation : relu
Test Loss : 1.4871
Test Accuracy : 0.4647

Model 5
Hidden Layers : (512, 256, 128)
Activation : tanh
Test Loss : 1.811
Test Accuracy : 0.3398

Model 6
Hidden Layers : (512, 256, 128)
Activation : sigmoid
Test Loss : 1.6096
Test Accuracy : 0.424

Model 7
Hidden Layers : (256, 128, 64)
Activation : relu
Test Loss : 1.5529
Test Accuracy : 0.4463

Model 8
Hidden Layers : (256, 128, 64)
Activation : tanh
Test Loss : 1.8038
Test Accuracy : 0.3588

Model 9
Hidden Layers : (256, 128, 64)
Activation : sigmoid
Test Loss : 1.6499
Test Accuracy : 0.3984

Model 10
Hidden Layers : (128, 64, 32)
Activation : relu
Test Loss : 1.5425
Test Accuracy : 0.4494

Model 11
Hidden Layers : (128, 64, 32)
Activation : tanh
Test Loss : 1.714
Test Accuracy : 0.3859

Model 12
Hidden Layers : (128, 64, 32)
Activation : sigmoid
Test Loss : 1.7648
Test Accuracy : 0.3634

```
=====
                        BEST MODEL
=====
Hidden Layers : (1024, 512, 256)
Activation    : relu
Best Accuracy : 0.4761
```

```
Predicted Label: Cat
Actual Label    : Cat
```

Predicted: Cat | Actual: Cat



In []: